VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY (VESIT)

Battery State Estimator: An ESN-Based Alternative to EKF and Coulomb Counting

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1. Problem & Motivation

• **Coulomb Counting:** Drifts rapidly over time due to current integration errors, sensor bias, and zero-point offset.
• **Extended Kalman Filter:** High accuracy but relies on exact physics and suffers parameter drift under thermal variation.
• **Deep RNNs/LSTMs:** Require excessive compute, high SRAM, and floating-point power beyond low-cost edge automotive MCUs.
• **Target:** Develop an edge-capable Leaky-Integrator ESN combined with EKF state observers and Verilog RTL for real-time BMS.

2. 2-RC Electro-Thermal Physics

• Models polarization voltages (V_1, V_2), ohmic resistance (R_0), and dynamic capacity fade.
• **Arrhenius Law:** Parameters dynamically scale with cell temperature:
$\theta(T) = \theta(T_{ref}) \cdot exp[(E_a / R_{gas})(1/T - 1/T_{ref})]$
• Dual-timescale decoupled SOC/SOH observer design.

3. Cyber-Physical Architecture

• **Physics Simulator Service:** Microservice simulating 2-RC cell dynamics, Arrhenius thermal variation, and fault modes.
• **Observer Pipeline:** Joint EKF, Unscented Kalman Filter (UKF), and Leaky-Integrator Echo State Network (ESN).
• **Leaky Reservoir Dynamics:**
$\tilde{x}_t = tanh(W_{in}[1; u_t] + W_{res}x_{t-1})$
$x_t = (1 - \alpha)x_{t-1} + \alpha \tilde{x}_t$
• Output trained via Ridge Regression (λ regularization).

4. Edge Embedded C99 Optimization

• **Compressed Sparse Row (CSR):** Stores only non-zero reservoir weights in row_ptr, col_idx, and val arrays.
• **Q15 Fixed-Point Math:** Replaces expensive float32 multiply-accumulate with 16-bit integer operations.
• **256-Entry LUT Tanh:** Piecewise linear interpolation eliminates floating-point transcendental functions.

5. Xilinx Artix-7 FPGA RTL

• **5-Stage Pipelined Architecture:** MAC → BIAS → SUM → TANH_IN → TANH_OUT.
• **Q6.10 Quantization:** 16-bit word length with 10 fractional bits for exact hardware-software parity.
• **BRAM Dual-Port Storage:** Input vector and state vector buffered in block RAM.
• **Bit-Exact Match:** 200/200 reservoir states match Python golden reference bit-for-bit (0 error).
• Target Device: Xilinx Artix-7 (xc7a100tcsg324-1) with 64-step NASA B0005 validation.

6. Comprehensive Test Suite

• **114 Pytest Tests:** 100% pass across simulator, chemistry, ESN model, estimator pipeline, and Verilog RTL.
• **Regression Runner:** Automated 8-step verification on Windows (`.bat`) and Linux/macOS (`.sh`).

7. Quantitative Results

Key Hardware & Software Metrics:
• **Execution Speedup: 6.7× faster** on ARM Cortex-M.
• **Flash Memory Savings: ~10 KB saved** via CSR SpMV.
• **Thermal Safety Accuracy: 98.40%** (Q15 fixed-point).
• **SOC Estimation RMSE: < 0.80%** (0.00803).
• **SOH Estimation RMSE: 5.73%** under aging cycles.

8. Live Demonstrations

• **Visualiser Dashboard:**
battery-visualizer.onrender.com
• **Physics Simulator:**
battery-physics-simulator.onrender.com
• **GitHub Release v2.0.0:**
github.com/tanish-jain-225/Battery_State_Estimator